Semiconductor Device-Level Modeling

SEMPY01 -Semiconductor Physics

AY 2025-2026 T1

Format

- This is an **individual assessment**.
- You must implement and analyze **all four canonical devices**:
  1. PN junction diode
  2. Bipolar junction transistor (BJT)
  3. MOSFET
  4. LED or photodiode
- Device modeling must use material parameters drawn from Assessments 1 and 2.
- Code must be your own, with clear plots and labeled axes.

Part A — PN Junction Diode

- Implement the diode equation:

$$I(V) = I_s \left(e^{qV/k_BT} - 1 \right),$$

  where I_s depends on n_i, diffusion coefficients, and diffusion lengths.
- Plot the diode I–V curve over forward and reverse bias.
- Compare behavior against a silicon diode baseline.

Part B — Bipolar Junction Transistor (BJT)

- Model a common-emitter configuration.
- Show how emitter/base/collector doping asymmetry affects current gain β.
- Plot I_C vs V_{CE} for multiple fixed I_B.
- Discuss how your semiconductor's E_g, n_i, and mobility affect transistor performance.

Part C — MOSFET

- Derive and implement drain current equations for the linear and saturation regions.
- Plot I_D vs V_{DS} for at least three different V_{GS} values.
- Comment on threshold voltage and channel mobility in your chosen material.

Part D — LED or Photodiode

- **LED case**: use E_g to estimate emission wavelength; calculate recombination rate and photon output.
- **Photodiode case**: model photocurrent vs. photon energy; show spectral cutoff at E_g.
- Include one literature band diagram or spectral curve for comparison.

Reflection and Discrepancy Analysis

Each section must include a short reflection:

1. What did you expect?
2. What did you find?
3. If there was a mismatch, why?

Grading Breakdown

- PN junction diode: **25%**
- BJT: **25%**
- MOSFET: **25%**
- LED/Photodiode: **15%**
- Reflection & discrepancy analysis: **10%**